

validations

DEFAULT_VALIDATIONS module-attribute

```
DEFAULT_VALIDATIONS = [VAL_NEGATIVE_AMOUNT, VAL_ZERO_AMOUNT, VAL_HIGH_AMOUNT, VAL_FUTURE_DATETIME, VAL_PREHISTORIC_DATETIME]
```

VAL_FUTURE_DATETIME module-attribute

```
VAL_FUTURE_DATETIME = Validation(desc='datetime in future', semantic=TIMESTAMP, sql_builder=lambda c: count(when(c > current_timestamp(), lit(True))))
```

VAL_HIGH_AMOUNT module-attribute

```
VAL_HIGH_AMOUNT = Validation(desc='amount > 1e8', semantic=AMOUNT, sql_builder=lambda c: count(when(isNotNull() & ~isnan(c) & (c > 100000000.0), lit(True))))
```

VAL_NEGATIVE_AMOUNT module-attribute

```
VAL_NEGATIVE_AMOUNT = Validation(desc='amount < 0', semantic=AMOUNT, sql_builder=lambda c: count(when(isNotNull() & (c < 0), lit(True))))
```

VAL_PREHISTORIC_DATETIME module-attribute

```
VAL_PREHISTORIC_DATETIME = Validation(desc='ancient transaction (<1950)', semantic=TIMESTAMP, sql_builder=lambda c: count(when(isNotNull() & (c < lit(datetime(1950, 1, 1))), lit(True))))
```

VAL_ZERO_AMOUNT module-attribute

```
VAL_ZERO_AMOUNT = Validation(desc='amount == 0', semantic=AMOUNT, sql_builder=lambda c: count(when(isNotNull() & (c == 0), lit(True))))
```

Semantics

Bases: `object`

Set of pre-defined semantics.

AMOUNT class-attribute instance-attribute

```
AMOUNT = Tag('AMOUNT', NUMERICAL)
```

TARGET class-attribute instance-attribute

```
TARGET = Tag('TARGET', CATEGORICAL)
```

TIMESTAMP class-attribute instance-attribute

```
TIMESTAMP = Tag('TIMESTAMP', DATETIME)
```

Tag [↗](#)

```
Tag(desc, ftype)
```

Bases: `object`

A Tag or semantic consist of a descriptive tag (desc) and a field type (ftype) associated to it.

desc `instance-attribute` [↗](#)

```
desc = desc
```

ftype `instance-attribute` [↗](#)

```
ftype = ftype
```

Validation [↗](#)

```
Validation(desc: str, semantic: Tag, sql_builder: Callable[[Column], Column], warning_fn: Callable[[int], bool] = lambda x: x > 0)
```

Bases: `object`

Class that represents a descriptive validation to be computed. These validations will be computed in a map and reduce approach.

Parameters:

Name	Type	Description	Default
desc ↗	<code>str</code>	the description/name of the validation.	<i>required</i>
semantic ↗	Tag	the semantic type to be applied to, e.g. Tag("AMOUNT", FieldType.NUMERICAL).	<i>required</i>
sql_builder ↗	<code>Callable[[Column], Column]</code>	Factory that receives a Spark <code>Column</code> and returns the Spark SQL validation expression	<i>required</i>
warning_fn ↗	<code>Callable[[int], bool]</code>	the function to be used to asses when a validation should be alerted.	<code>lambda x: x > 0</code>

desc `instance-attribute` [↗](#)

```
desc = desc
```

semantic `instance-attribute` [↗](#)

```
semantic = semantic
```

sql_builder `instance-attribute` [↗](#)

```
sql_builder = sql_builder
```

warning_fn instance-attribute [¶](#)

```
warning_fn = warning_fn
```

build_validations_dict [¶](#)

```
build_validations_dict( semantics_map: dict[str, Tag], validations: Any | None = None) -> dict[str, list[Validation]]
```

Helper function that builds a dict with the validations to be performed per field.

Parameters:

Name	Type	Description	Default
semantics_map ¶	dict[str, Tag]	dictionary matching fields (keys) to semantic tags (values).	<i>required</i>
validations ¶	Any None	set/list/tuple containing list of Validation.	None

Returns:

Type	Description
unknown	dictionary with list of validations (values) associated to each field (keys).

check_validations_type [¶](#)

```
check_validations_type(validations, ftype, fname='')
```

Helper function to check if the list of validations matches the expected field type, otherwise it raises an exception.

Parameters:

Name	Type	Description	Default
validations ¶		list of Validation.	<i>required</i>
ftype ¶		FieldType.	<i>required</i>
fname ¶		(optional) field name.	''

show_validations_by_field [¶](#)

```
show_validations_by_field(semantics_map, validations=None)
```

Helper function that prints the validations to be performed per field given a semantics map and a list of validations.

Parameters:

Name	Type	Description	Default
semantics_map ¶		dictionary matching fields (keys) to semantic tags (values).	<i>required</i>

Name	Type	Description	Default
<code>validations</code> 		set/list/tuple containing list of Validation.	<code>None</code>